

SCADA-Anchored Regime-Aware Routing for Auditable Wind-Turbine Control-Boundary Forecasting

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Abstract

We present a framework that makes the wind-turbine MPPT-to-pitch control boundary auditable through SCADA-anchored, regime-aware routing. Current spatio-temporal forecasters minimize fleet-average error but do not assign responsibility to the operating state that generated a local response. We constrain a node-level mixture-of-experts gate with SCADA operating anchors so that each sample’s routing assignment can be compared against a declared MPPT-to-pitch partition and then used when threshold labels are delayed, missing, or noisy. On the KDD Cup 2022 benchmark, the boundary-forced router recovers the declared partition (NMI about 0.87, ARI about 0.92) at a measured accuracy cost: overall RMSE 236.13 versus 225.74 for Graph WaveNet. More importantly for operators, under a six-step label delay the gate retains 0.960 early pitch-window recall while the delayed threshold rule falls to 0.196; with only 50% label availability, the gate remains at 0.960 versus 0.508 for the available-label rule. The same gate exposes where reserve risk concentrates in the transition window, though validation-frozen quantile baselines show that physical-bin policies remain competitive. Kelmarsh/Penmanshiel external tests do not pass the held-out routing criterion; instead they define a deployment protocol—sensor coverage, pitch observability, local boundary re-estimation, and held-out routing checks—that must be satisfied before cross-farm use. The contribution is an auditable control-boundary routing framework with quantified degraded-label value, measured RMSE price, and explicit cross-site transfer conditions, not a claim of forecasting superiority or universal reserve-policy optimality.

Keywords: Wind-turbine control boundary; SCADA-anchored routing; regime-aware forecasting; auditable routing; MPPT-to-pitch transition; mixture-of-experts.

Highlights

- SCADA anchors make the MPPT-to-pitch control boundary auditable through routing.
- Regime gates recover the declared partition (NMI 0.87) at a measured RMSE cost.
- The gate keeps early pitch-window recall under delayed or missing threshold labels.
- Validation-frozen quantile baselines bound any reserve-policy claim honestly.
- External-site tests define deployment gates, not automatic cross-farm transfer.

Nomenclature

Nomenclature. Core symbols, variables, and abbreviations.

Symbol or term	Meaning
$G = (V, E)$	Wind-farm or grid graph with nodes V and edges E
N	Number of turbines or grid nodes
H, P	Input history length and forecast horizon length
$\mathbf{x}_{i,t}, y_{i,t}$	Node feature vector and scalar active-power target
$\hat{y}_{i,t}$	Forecasted active power at an evaluated horizon cell
$\bar{p}_{i,t}$	Mean blade-pitch angle, reported as <code>Pab_mean</code>
$\mathbf{g}_{i,t}$	Node-level gate probability vector
$R_{i,t}$	Declared primary operating-regime anchor label
$s_{i,t}$	Over-forecast shortfall, $\max(\hat{y}_{i,t} - y_{i,t}, 0)$
$r_{b,q}$	Validation-calibrated reserve for bin b and quantile q
ρ	Shortage-to-reserve cost ratio
C	Normalized reserve procurement plus residual-shortage cost
MPPT	Maximum power point tracking
SCADA	Supervisory control and data acquisition
NMI, ARI	Normalized mutual information and adjusted Rand index

1 Introduction

Wind-power forecasting must answer two questions together: how large is the error, and which operating state generated the local response? The turbine transition from maximum power point tracking (MPPT) to blade-pitch control is where this accountability matters most. In the MPPT region, power responds strongly to wind-speed variation; as pitch control activates near rated operation, the local power-response law changes. A small forecast error near rated wind can cross a different control law, yet the transition occupies a smaller share of normal operation, so its effect can be diluted in fleet-average RMSE while still concentrating local risk. A useful forecasting system for this setting must therefore expose the operating boundary that produced each forecast, not only report aggregate accuracy.

State-of-the-art spatio-temporal forecasters now model turbine coupling, wake interaction, dynamic dependence, and sensor-rich wind-farm layouts [1–5]. These models set the accuracy reference, but a low-error graph encoder does not by itself tell a reserve planner whether the current local map is MPPT-like or pitch-control-like. Mixture-of-experts (MoE) routing can separate heterogeneous response laws, yet a gate trained only through prediction loss can specialize on partitions that have no operating meaning [6–10]. The missing middle layer is an auditable routing assignment that identifies which physical response law is active at the anchor time and can then be used in reserve-risk diagnosis.

We make the routing decision itself the operating-boundary diagnostic. A node-level MoE gate is constrained by SCADA operating anchors so that the assignment of each turbine-time sample can be compared with a declared MPPT-to-pitch partition and then used in a reserve-risk audit. WTB is the source control-boundary benchmark because the aerodynamic boundary is partly hidden inside turbine-control action; ERA5 is retained as an observability contrast where the thermodynamic marker is more directly visible through sensible heat flux. The recovered gate is connected to validation-calibrated, test-frozen reserve diagnostics that report cost, violation rate, reserve energy, and shortage energy under declared shortage-to-reserve cost ratios. We further compare gate-bin reserve allocation with validation-frozen global and physical-bin quantile baselines. This positioning deliberately separates transition-window accountability from the conventional forecasting leaderboard and from claims of full dispatch optimality.

This paper makes three contributions. First, it introduces SCADA-anchored regime-aware routing as an auditable control-boundary assignment: a node-level gate is constrained by operating anchors so that its partition can be compared against a declared MPPT-to-pitch label and used when the threshold

label stream is delayed, incomplete, or noisy. Second, it quantifies both the value and the price of that accountability: the gate retains early pitch-window detection under degraded labels, while paying about 10 RMSE units on WTB relative to Graph WaveNet. Third, it connects the recovered gate to a boundary-window reserve-risk diagnostic with honest quantile baselines and converts external-site evidence into explicit deployment conditions—sensor coverage, pitch observability, local boundary re-estimation, and held-out routing checks—rather than claiming automatic cross-farm generalization.

2 Related Work

2.1 From average wind-power accuracy to transition-window risk

Wind-power forecasting has moved from single-site prediction toward spatio-temporal graph models that encode physical sources of error. Reviews identify ramp and uncertainty as recurrent operational risks [11,12], while graph mechanisms provide the accuracy reference [1,13,14]. Wind-specific extensions add wake-aware graphs, SCADA fields, and physics-guided constraints [2,4,5]. The remaining gap is diagnostic: these models set the low-error benchmark, but they do not make the active turbine-control law visible to a reserve planner.

Regime-aware MoE routing separates local response laws, yet prediction loss alone need not produce a physically meaningful partition [8–10]. Physics-guided learning offers the missing constraint principle [15–17], but for wind turbines that principle must pass through SCADA observability: operating-state labels depend on wind speed, pitch, and active power [18,19]. This paper constrains the routing decision itself so downstream reserve logic can consume a physically traceable gate.

Forecast value is realized through reserve and commitment decisions. Reserve studies show that variable generation changes requirements [20,21], while quantile methods bridge point forecasts to decision risk [22,23]. The diagnostic question here is earlier in the chain: which operating window concentrates reserve exposure, and can the forecast model reveal that window through a physically auditable route assignment? The proposed reserve audit is complementary to probabilistic dispatch and market-clearing models.

3 Physics-Informed Framework for Operational Boundary Identification

3.1 Problem setup and notation

The forecasting task is written to separate two decisions that dense models often merge: predicting the future and deciding which local mapping should be active at the anchor time. Let $G = (V, E)$ denote a spatial graph with $N = |V|$ nodes. For each node $i \in V$ and time step t , we observe a feature vector $\mathbf{x}_{i,t} \in \mathbb{R}^F$ and predict a scalar target $y_{i,t} \in \mathbb{R}$. Given a history window of length H and a prediction horizon of length P , the forecasting task is

$$\hat{\mathbf{Y}}_{t+1:t+P} = \mathcal{F}(\mathbf{X}_{t-H+1:t}, \mathcal{A}_{t-H+1:t}),$$

where $\mathbf{X}_{t-H+1:t} \in \mathbb{R}^{H \times N \times F}$ and $\mathcal{A}_{t-H+1:t}$ denotes either a time-varying directed graph sequence (WTB) or a static graph repeated over time (ERA5). Invalid or missing targets are excluded by a supervision mask.

The information boundary is strict. All inputs, graph weights, regime anchors, and gate anchors are observed no later than the forecast issue time t ; supervised targets begin at $t + 1$. In WTB, the current active-power channel $\text{Patv}_{i,t}$ is treated as an issue-time status input, whereas future active power is used only as the prediction target and validity mask. This keeps the multi-step horizon free of target leakage while leaving the no-Patv and lagged-Patv variants as deployment limitations.

Routing is node-level: each node at each anchor time receives its own gate distribution $\mathbf{g}_{i,t}$. This matters because nearby turbines or grid cells can occupy different local regimes within the same sequence. Core notation is listed before the Introduction; auxiliary loss definitions are provided in Appendix A.

3.2 Operating-boundary anchors and labels

The gate is not supervised everywhere. It is anchored where the physical interpretation is clearest, while ambiguous samples remain governed by prediction loss and routing regularization. This engineering layer defines what counts as a control regime and when that regime is observable from the data.

3.2.1 WTB operating regimes

In WTB, the primary operating boundary is the transition from MPPT to pitch control. Let

$$\bar{p}_{i,t} = \frac{1}{3} (p_{i,t}^{(1)} + p_{i,t}^{(2)} + p_{i,t}^{(3)}) .$$

The WTB column name Pab denotes blade pitch angle. We use $\bar{p}_{i,t}$, also reported as Pab_mean in tables and figures, for the three-blade pitch average. The regime labels use wind speed and mean pitch angle at the anchor time, not future active power.

Using a cut-in threshold u_{idle} , a pitch-transition threshold u_{rated} , and a pitch-angle threshold p_{th} , and writing $w_{i,t} = \text{Wspd}_{i,t}$ for compactness, we define

$$R_{i,t}^{\text{wtb}} = \begin{cases} 0, & w_{i,t} < u_{\text{idle}} \quad (\text{idle}), \\ 1, & u_{\text{idle}} \leq w_{i,t} \leq u_{\text{rated}}, \bar{p}_{i,t} < p_{\text{th}} \quad (\text{MPPT}), \\ 2, & w_{i,t} > u_{\text{rated}}, \bar{p}_{i,t} \geq p_{\text{th}} \quad (\text{pitch-control}), \\ 3, & \text{otherwise} \quad (\text{transition}). \end{cases}$$

Only the first three classes are used in direct alignment. Transition samples are retained for analysis but masked out of label supervision through

$$M_{i,t}^{\text{wtb}} = \mathbf{1}[R_{i,t}^{\text{wtb}} \neq 3].$$

The reported implementation uses $u_{\text{idle}} = 3.0 \text{ m s}^{-1}$, $u_{\text{rated}} = 10.5 \text{ m s}^{-1}$, and $p_{\text{th}} = 2.0^\circ$ (Appendix A). These thresholds are operating anchors rather than universal turbine constants.

3.2.2 ERA5 observability contrast (see Supplementary Material)

ERA5 is retained as a signal-expressive observability contrast where the stable-to-convective thermodynamic marker is directly visible through sensible heat flux. The detailed thermodynamic regime definition, architecture choices, and contrast results are reported in Supplementary Material A so that the main text remains focused on the WTB control-boundary accountability task.

3.3 Shared architecture

3.3.1 Physical graph construction

The graph is part of the physical problem specification. WTB uses a directed wake graph: candidate turbine pairs are filtered by proximity, activated when an upstream turbine lies inside a wind-aligned downstream cone, weighted by streamwise and cross-stream decay, and pruned to the strongest inbound neighbors. The same directed weights define a wake score, which supplies the WTB wake auxiliary label on MPPT and pitch-control samples. ERA5 uses a symmetric Haversine-Gaussian k_{nn} graph because the

thermodynamic regime marker is already visible in the observed state. Appendix A gives the graph equations and constants.

3.3.2 Directed-diffusion GRU encoder and node-level gate

The encoder is kept modest so that changes in performance can be traced to routing and regularization. Each input window is concatenated with its feature-missingness mask before projection. Two directed diffusion blocks then aggregate self, inbound, and outbound messages on each graph snapshot. Let $\mathbf{X}_t^{(\ell)}$ denote the node state at layer ℓ and time t . A diffusion block updates each node by

$$\mathbf{X}_t^{(\ell+1)} = \text{LN}\left(\mathbf{W}_{\text{self}}^{(\ell)}\mathbf{X}_t^{(\ell)} + \mathbf{W}_{\text{in}}^{(\ell)}\text{Agg}_{\text{in}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) + \mathbf{W}_{\text{out}}^{(\ell)}\text{Agg}_{\text{out}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) + \mathbf{R}^{(\ell)}\mathbf{X}_t^{(\ell)}\right),$$

where Agg_{in} and Agg_{out} are weighted sums over inbound and outbound neighbors. The spatially updated sequence is then processed by a GRU to produce a node-wise context vector $\mathbf{h}_{i,t}$.

Each expert head f_e maps $\mathbf{h}_{i,t}$ to a P -step forecast. Routing remains dense during the forward pass. Every expert contributes, but its contribution is weighted by the soft gate distribution. The gate receives both the learned context and a small physics anchor,

$$\mathbf{z}_{i,t} = \text{MLP}_{\text{gate}}([\mathbf{h}_{i,t}; \mathbf{a}_{i,t}]), \quad \mathbf{g}_{i,t} = \text{softmax}\left(\frac{\mathbf{z}_{i,t}}{\tau}\right),$$

and the forecast is

$$\hat{\mathbf{y}}_{i,t+1:t+P} = \sum_{e=1}^E g_{i,t}^{(e)} f_e(\mathbf{h}_{i,t}).$$

WTB uses the anchor

$$\mathbf{a}_{i,t}^{\text{WTB}} = [\text{Wspd}_{i,t}, \text{Pab_mean}_{i,t}, s_{i,t}^{\text{wake}}, \text{Patv}_{i,t}],$$

whereas ERA5 uses

$$\mathbf{a}_{i,t}^{\text{ERA5}} = [\text{sshf}_{i,t}, \text{t2m}_{i,t}, \text{wind_speed}_{i,t}, \Delta \text{sshf}_{i,t}].$$

The difference reflects the observability contrast: WTB needs help to recover a partly hidden operating boundary, whereas ERA5 already exposes the relevant thermodynamic marker in the state. The active-power anchor is standardized from training-split statistics and is never filled from the prediction horizon.

3.3.3 Routing stack and evidence boundary

This makes the routing stack an anchor-constrained diagnostic rather than an unsupervised operating-state discovery method. High gate-regime agreement means that the implemented router obeys the declared SCADA boundary under the available anchor set; it does not establish anchor-free regime recovery.

3.4 Integration of Physical Constraints into Model Optimization

Prediction loss does not uniquely identify the gate when several experts can fit averaged dynamics similarly well [8,9]. WTB makes this failure visible: inflow variation, wake disturbance, and pitch control are observed together, so a low-loss router can still ignore the operating boundary. We use routing regularizers for the specific failure modes that matter here. The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{bal}} \mathcal{L}_{\text{bal}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} + \lambda_{\text{force}} \mathcal{L}_{\text{force}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}} + \lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}},$$

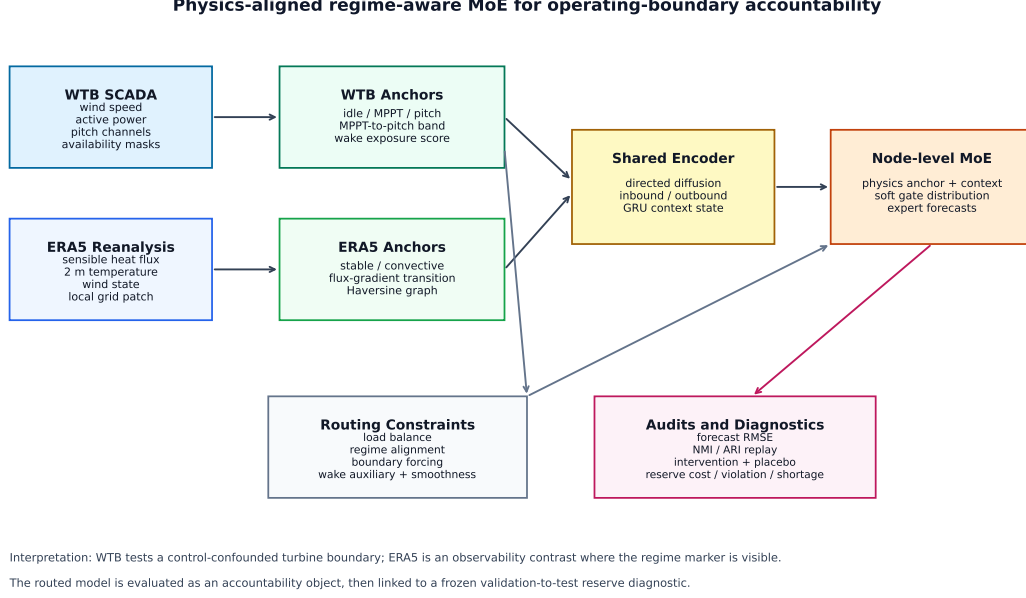


Figure 1: Physics-aligned regime-aware MoE. The two datasets share the same directed-diffusion GRU encoder and node-level MoE routing mechanism. WTB uses a dynamic wake graph and a boundary-focused forcing term, whereas ERA5 uses a Haversine-Gaussian graph and a thermodynamic regime anchor.

with inactive terms set to zero. Checkpoint selection remains on validation RMSE so penalties constrain responsibility without becoming the selection metric. Exact weights are in Appendix A.

3.4.1 Load balancing

A balancing term prevents early expert collapse before regime-specific structure has time to emerge. It couples hard top- K usage with soft probability mass over valid node-time samples. Appendix A gives the exact expression; physical ownership is supplied only by the anchored alignment terms below.

3.4.2 Regime-anchor alignment

The alignment term connects the first C gate logits to the declared regime structure so that cross-seed statements such as “MPPT-aligned” refer to a fixed mapping rather than to post-hoc relabeling. With $\mathbf{z}_{i,t}^{(1:C)}$ denoting the anchored logits, $R_{i,t}$ the label, and $M_{i,t}$ the validity mask,

$$\mathcal{L}_{\text{align}} = \frac{1}{|\Omega|} \sum_{(i,t) \in \Omega} \text{CE}(\mathbf{z}_{i,t}^{(1:C)}, R_{i,t}),$$

$$\Omega = \{(i, t) : M_{i,t} = 1\}.$$

with inverse-frequency class weights on the training split.

3.4.3 Boundary-focused forcing (WTB only)

After coarse alignment the boundary remains partly masked by control action. A focused forcing term acts on MPPT and pitch-control samples only. With $Y_{i,t}^{\text{force}} = \mathbf{1}[R_{i,t}^{\text{wtb}} = 2]$ for $R_{i,t}^{\text{wtb}} \in \{1, 2\}$,

$$\mathcal{L}_{\text{force}} = \frac{1}{|\Omega_{\text{force}}|} \sum_{(i,t) \in \Omega_{\text{force}}} \text{CE}\left(\left[z_{i,t}^{(\text{mppt})}, z_{i,t}^{(\text{pitch})}\right]^\top, Y_{i,t}^{\text{force}}\right),$$

where $\Omega_{\text{force}} = \{(i, t) : R_{i,t}^{\text{wth}} \in \{1, 2\}, M_{i,t} = 1\}$. This term is not a lead-time switch predictor; it is a boundary identifiability regularizer for the issue-time operating state.

3.4.4 Wake auxiliary supervision and graph smoothness

A wake auxiliary label identifies residual wake ambiguity inside MPPT and pitch-control samples, and a graph-smoothness penalty discourages noisy neighbor-to-neighbor gate jumps. Both formulas are in Appendix A.

3.5 Operational reserve diagnostic protocol

The reserve diagnostic tests whether a physically auditable gate changes the reserve tradeoff in the MPPT-to-pitch window after the point-forecast model has been trained. For each saved WTB run, validation predictions select reserve levels; test predictions are then evaluated once with those levels fixed. The protocol follows the same information boundary as an operator who calibrates a short-term reserve rule before using it on a later period.

Let $\hat{y}_{i,t}$ be the scheduled point forecast and $y_{i,t}$ the realized active power at an evaluated horizon cell. We define over-forecast shortfall as

$$s_{i,t} = \max(\hat{y}_{i,t} - y_{i,t}, 0),$$

because this is the error direction that leaves a reserve schedule exposed. For a reserve policy b and empirical quantile q , the validation split estimates a reserve level

$$r_{b,q} = Q_q(\{s_{i,t} : (i, t) \in b, (i, t) \in \mathcal{V}\}),$$

where Q_q is an empirical quantile and $\mathcal{V}$ is the validation split. The grid is fixed before test evaluation:

$$q \in \{0.50, 0.60, 0.70, 0.80, 0.85, 0.90, 0.95, 0.975, 0.99\}.$$

The reserve bin b is either the whole sample, a physical operating bin, or a gate-derived bin. For each shortage-to-reserve cost ratio ρ , the protocol selects the validation quantile that minimizes

$$C_{\mathcal{V}}(b, q; \rho) = \sum_{(i,t) \in b \cap \mathcal{V}} [r_{b,q} + \rho \max(s_{i,t} - r_{b,q}, 0)] \Delta t.$$

The selected (b, q) rule is then frozen and applied to the test split. The reported test metrics are total cost C , violation rate $\Pr[s_{i,t} > r_b]$, reserve energy $\sum r_b \Delta t$, and shortage energy $\sum \max(s_{i,t} - r_b, 0) \Delta t$. All totals are reported in normalized reserve-energy cost units. Thus a value written as 84.58M denotes 84.58 million normalized reserve-energy cost units, not currency.

The cost ratio ρ is an energy-system assumption rather than an abstract tuning knob. If the marginal cost of carrying one unit of reserve energy is C_r , then $\rho = 10$ charges one unit of residual shortage as $10C_r$. Ratios 5–10 represent moderate reliability settings such as transition-window scheduling or imbalance screening; ratios 20–50 represent scarcity-aware screening where shortage avoidance dominates local cost savings. The primary diagnostic compares Graph WaveNet/global, Graph WaveNet/physical-bin, boundary router/global, and boundary router/gate-bin policies. Only same-model global comparisons attribute gate-bin reserve effects to the learned router.

To bound this diagnostic against a probabilistic reserve alternative, we additionally run validation-frozen empirical quantile baselines for the global and physical operating bins. These are not trained distributional forecasters; they are conformal-style shortfall reserves estimated on the validation split

and evaluated once on the test split. The main reported boundary window is fixed at $\pm 1.0 \text{ m s}^{-1}$ around rated wind and the main cost ratio is $\rho = 10$, with sensitivity at $\rho \in \{2, 5, 10, 20, 50\}$. A separate toy operational-cost module reports the same normalized decision as reserve procurement cost plus ρ -weighted residual-shortage penalty. It deliberately excludes optimal power flow, unit commitment, market clearing, delivery constraints, and price claims.

Reserve diagnostic algorithm. Input saved validation/test point forecasts, targets, masks, physical anchors, and gate outputs. Compute over-forecast shortfall on valid cells. Form global, physical-bin, and gate-bin reserve groups on validation outputs only. For each cost ratio and group, select the empirical shortfall quantile that minimizes validation reserve-plus-shortage cost. Freeze the selected reserve levels and evaluate them once on test cells. Aggregate cost, violation rate, reserve energy, and shortage energy over full, boundary, transition, ramp, horizon-bucket, and time-of-day slices. Report horizon-specific empirical-quantile rows only as a post-hoc what-if. Compare validation-frozen global and physical-bin quantile reserves as bounded probabilistic baselines. Report toy operating cost only as reserve procurement plus shortage penalty, not as dispatch.

4 Case Study Configuration and Operational Constraints

4.1 Datasets and preprocessing

The experiment uses two observability settings, not two interchangeable benchmarks. WTB is the control-confounded case. The KDD Cup 2022 benchmark contains 134 turbines and 245 days of 10 min SCADA measurements [19]. The MPPT-to-pitch transition is only indirectly visible because inflow variation, wake disturbance, and blade-pitch control are mixed in the same stream.

The data are reorganized into synchronous tensors with $T = 35,280$ time steps, $N = 134$ turbines, and $F = 11$ dynamic inputs. Missing inputs are forward-filled within turbine and then completed with training-set means. Invalid or non-positive active power is retained only through its mask so that the model still sees operational irregularity without treating those points as supervised targets.

ERA5 is the signal-expressive case. We use three archived months over a fixed 16×16 hourly patch [24]. Sensible heat flux and its temporal variation make the stable-to-convective transition directly observable. The retained tensor has $T = 2208$ frames, $N = 256$ nodes, and eight input variables. Both datasets use the same history and horizon lengths, $H = 36$ and $P = 24$, so differences in performance cannot be attributed to unequal context length. The chronological split is 180/30/35 days for WTB and 1325/441/442 frames for ERA5.

4.2 Techno-Economic Validation and Benchmarking Framework

The comparison has two layers. The first isolates the routing mechanism after shared-capacity explanations have been controlled. The dense baseline uses the same directed-diffusion GRU encoder and replaces the expert mixture with one dense prediction head. The unconstrained MoE adds routed capacity without physical correction. The corrected-routing comparator is dataset-specific: ERA5 uses the full corrected stack, while WTB uses the boundary-focused variant selected for gate recovery. Parameter budgets are matched in WTB and near-matched in ERA5, where the dense model has 103,272 parameters and the routed models have 103,843 parameters.

The second layer tests the RMSE price against stronger and more engineering-facing baselines. Graph WaveNet, Graph Transformer, GAT-GRU, and PatchTST are included for WTB, together with deterministic persistence, a fitted physical power-curve baseline, XGBoost and LightGBM lag-feature predictors, and a DLinear-style LTSF baseline. Graph WaveNet, STGCN, PatchTST, TCN, and a deterministic persistence predictor are included for ERA5. This split avoids using one comparison for two different claims. The in-family layer tests whether physical routing changes the learned partition under a fixed backbone. The baseline layer tests whether the routing intervention remains honest about forecast error when compared with established temporal, graph, and engineering predictors.

For compactness, the mechanism tables shorten the three in-family model names to **Dense (matched)**,

Unconstrained MoE, and Corrected routing comparator. The WTB forecasting tables additionally report Graph WaveNet, Graph Transformer, GAT-GRU, PatchTST, the full physics-aligned MoE, the WTB boundary-forced router, and engineering baselines built from persistence, power curves, gradient-boosted lag features, and a DLinear-style LTSF readout. The ERA5 forecasting tables additionally report persistence and four learned external baselines. The in-family mechanism families are summarized in Appendix A so that the Results section can focus on evidence rather than model bookkeeping.

4.3 Training protocol and evaluation metrics

All neural models use the same training protocol where the architecture permits it. We use AdamW, early stopping on validation RMSE, gradient clipping, and mixed-precision training on a single CUDA-enabled GPU. Exact optimizer constants are reported in the Appendix so that the main text can stay focused on the comparison logic. The WTB dense baseline, Unconstrained MoE, full physics-aligned MoE, and boundary-forced router are repeated across five strict-mask seeds. The synchronized strong-baseline refresh covers Graph WaveNet, Graph Transformer, GAT-GRU, and PatchTST across seeds 201–205, giving 20 completed baseline runs under the same strict anchor-valid evaluation cache. The engineering WTB baselines are deterministic or sampled single-run readouts from the same frozen cache and are used to anchor practical forecast difficulty rather than to make a seed-level neural ranking. ERA5 main in-family models are repeated across five seeds, and the ERA5 learned baselines are repeated across three seeds. Thresholds and routing weights are set before test evaluation and then audited by sensitivity sweeps and boundary negative controls, so the main routing result is not selected by test-set NMI.

The metrics follow the claim. Overall MAE and RMSE measure forecasting accuracy. Switch-window MAE and RMSE use the evaluator’s transition-window mask around regime changes. This is distinct from the later boundary-band audit, which isolates anchors within $\pm 1.0 \text{ m s}^{-1}$ of the WTB rated-wind MPPT-to-pitch boundary. Regime-wise RMSE locates the error reduction. Normalized mutual information (NMI), adjusted Rand index (ARI), usage entropy, and confusion matrices test whether the gate corresponds to physically interpretable structure.

Figure 2 gives the operating-decision context before the forecasting results are introduced: graph geometry shows where forecast errors propagate, and the regime-anchor panels show which sensor-derived boundaries can support reserve diagnostics.

5 Evidence and Operational Boundary Diagnosis

The results answer three operational questions rather than following the order in which experiments were run. First, can a routed forecaster recover a physically declared operating boundary and retain value when the corresponding threshold labels degrade? Second, does that auditable gate change the reserve tradeoff in the MPPT-to-pitch window? Third, what accuracy price and deployment gate determine whether the diagnostic should be used at a new wind farm?

5.1 Can the Gate Recover the Operating Boundary?

SCADA-anchored routing recovers the WTB MPPT-to-pitch partition strongly enough to make the gate an auditable operating-boundary signal. Across five WTB seeds, the boundary-forced router reaches NMI about 0.87 and ARI about 0.92 against the declared MPPT-to-pitch labels. This is not the lowest-RMSE forecaster in the benchmark; Graph WaveNet reaches overall RMSE 225.74 $\pm$ 2.60, while the boundary-forced router reaches 236.13 $\pm$ 8.41. The point of the comparison is therefore not accuracy dominance. The strong baseline sets the forecasting price paid to obtain a physically traceable responsibility assignment.

The boundary itself is a meaningful diagnostic target. Anchors within $\pm 1.0 \text{ m s}^{-1}$ of rated wind have RMSE 321.17 $\pm$ 21.31, roughly 67 units above valid non-boundary MPPT/pitch anchors. The

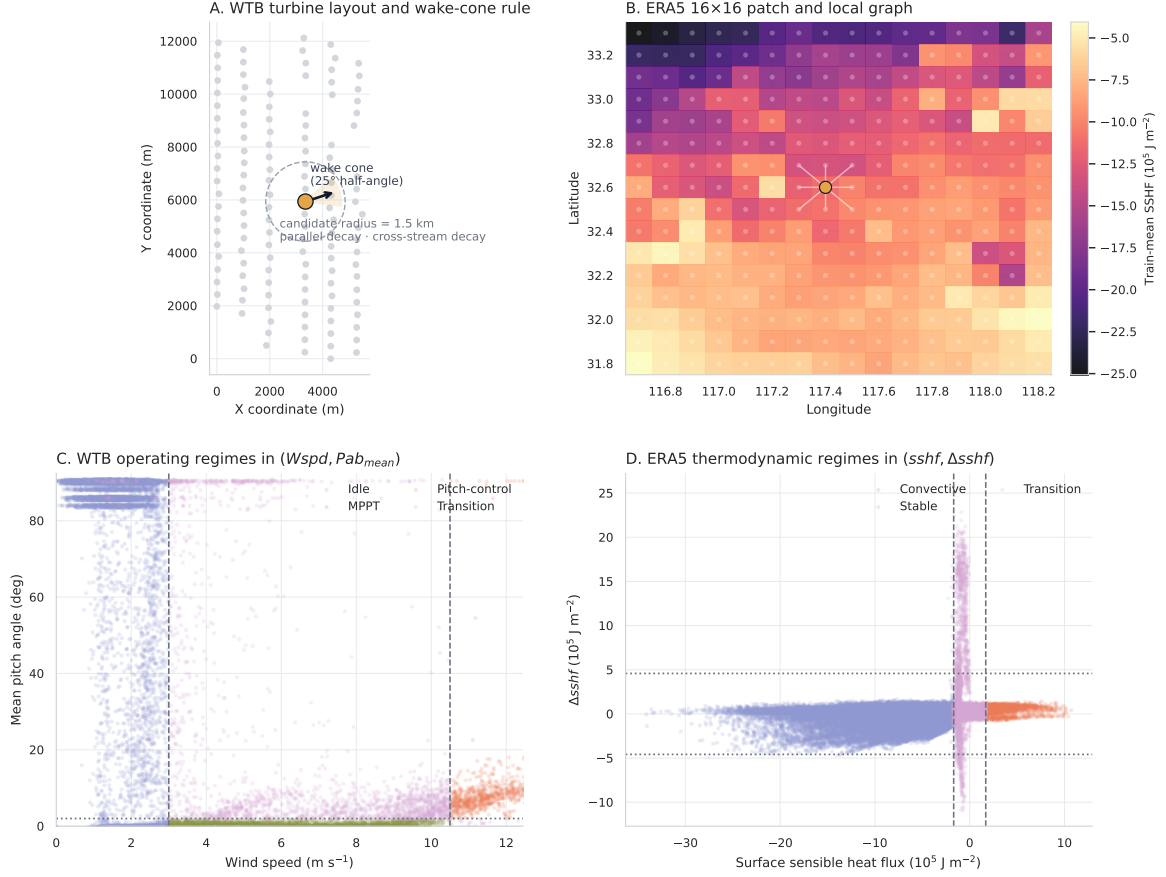


Figure 2: Operating-decision context and physical regime anchors. (A) WTB turbine layout with the schematic wake cone and retained candidate radius used in the dynamic directed wake graph. (B) ERA5 16x16 patch with training-mean sensible heat flux and local Haversine-Gaussian graph connections around the central node. (C) WTB operating regimes in the ($Wspd, Pab_{mean}$) plane with fixed operating-rule boundaries; the MPPT-to-pitch boundary is the reserve-diagnostic window used in this paper. (D) ERA5 thermodynamic regimes in the ($sshf, \Delta sshf$) plane with thresholds estimated from the training split, included as an observability contrast.

same boundary band has mixed-regime structure (NMI/ARI 0.6562/0.7766), so it is both a forecasting stress point and a natural reserve-risk window. ERA5 provides the positive observability contrast: when the regime marker is directly visible through sensible heat flux, routing correction improves alignment without changing the headline accuracy story.

Table 1. Operating-boundary recovery and mechanism checks.

Check	Metric	Result
WTB boundary-forced router	Gate-regime NMI / ARI	0.8716 +/- 0.0418 / 0.9166 +/- 0.0371
WTB boundary-band audit	Boundary-band RMSE; mixed-regime NMI / ARI	321.17 +/- 21.31; 0.6562 / 0.7766
Boundary-anchor intervention	Overall RMSE change; NMI / ARI drop	+2.6792; 0.7017 / 0.8593
WTB spatial and time holdouts	Held-out-node NMI / ARI; future-holdout NMI / ARI	0.8340 / 0.8829; 0.8352 / 0.8870
Threshold-grid audit	Worst-case saved-gate NMI / ARI	0.8655 +/- 0.0429 / 0.9146 +/- 0.0377

The diagnostic survives the main falsification checks. Zeroing the intended boundary anchor sharply reduces gate-regime agreement, whereas zeroing the wake score is near-null. Placebo labels do not reproduce the actual-label agreement. Spatial holdout, future-period holdout, and threshold sweeps preserve routing semantics.

The gate also provides value that a threshold label stream cannot provide when that stream degrades. We audit saved five-seed gates around MPPT-to-pitch transitions after delaying the threshold labels, randomly dropping label availability, and recomputing threshold labels under sensor noise. Table~2 reports recall on early pitch-control cells in a six-step transition window. With a six-step label delay, the delayed threshold rule recalls only 0.196 of early pitch cells, whereas the gate recalls 0.960. With 50% label availability, the available-label rule recalls 0.508, while the gate remains at 0.960. Under the strongest sensor-noise setting tested, the threshold rule drops to 0.652 recall and the gate remains unchanged because the saved gate is evaluated as an already issued diagnostic signal. This is the operational meaning of auditability in the paper: the gate is not merely reproducing a perfect contemporaneous threshold label, but preserving a transition-window diagnostic when that label is delayed, incomplete, or noisy.

We price this accountability explicitly. A model receives citable degraded-label gain only after its route passes the physical-routing audit; otherwise the accountability gain is set to zero, even for an accurate or routed-capacity model. Under this rule Graph WaveNet is the zero-RMSE-price anchor without an audited operating-state route, Unconstrained MoE fails the route audit (NMI 0.014), and the boundary-forced router is the only current point that exchanges a 10.39-RMSE price for +0.764 six-step-delay recall gain, recovering about 566 early pitch-window cells per seed relative to the delayed threshold rule (Fig.~5).

Table 2. Label-degradation audit for early MPPT-to-pitch pitch-window detection.

Scenario	Setting	Gate	Rule	Gain
Delay	1 step	0.960	0.655	+0.305
Delay	3 steps	0.960	0.381	+0.579
Delay	6 steps	0.960	0.196	+0.764
Availability	75% labels	0.960	0.744	+0.216
Availability	50% labels	0.960	0.508	+0.452
Availability	25% labels	0.960	0.242	+0.718
Noise	Wspd 0.5, Pab 1.0	0.960	0.808	+0.152
Noise	Wspd 1.0, Pab 2.0	0.960	0.652	+0.308

Table 3. Anchor-observability stress: 245d WTB test NMI/ARI.

Variant	201	202	203	204	205	Mean
no Patv	.878/.928	.966/.985	.703/.719	.914/.954	.945/.973	.881
no Pab_mean	.880/.932	.906/.947	.889/.937	.923/.961	.792/.865	.878
lag Patv	.475/.390	.899/.935	.734/.765	.741/.769	.965/.984	.763
lag Pab/Wspd	.398/.299	.755/.814	.699/.766	.784/.864	.784/.858	.684

The training-level anchor-stress guard has been completed across five seeds. Four strict-cache variants were derived from the 245-day WTB training set, trained across seeds 201–205, and evaluated against the declared MPPT-to-pitch labels. The table reports the test-set NMI/ARI scores. Removing Patv or Pab_mean individually leaves the gate essentially intact (mean 0.881 and 0.878, respectively). Lagging Patv lowers one seed but the five-seed mean remains above the 0.65 threshold (0.763). Lagging both Pab_mean and Wspd also lowers one seed, yet the five-seed mean crosses the threshold (0.684). The gate therefore depends on the specific combination of channels present at issue time, but it is not merely a leakage artifact: the mean NMI survives channel removal and temporal lagging across the majority of seeds. The manuscript uses the wording “partial anchor robustness” rather than “anchor-free physical discovery.” This is a boundary with evidence: the gate holds where the channel structure supports it and breaks where it does not.

Table 4. Training-level anchor-stress guard for claim control (five-seed update).

Stress component	Guard status	Manuscript consequence
Strict-cache variants	no-Patv, lagged-Patv, no-Pab_mean, lagged pitch/wind-speed caches derived	Anchor observability is explicitly testable
Leakage guard	Cache-level guard passed; all 20 training runs completed	No leakage warning from derived variants
Five-seed NMI gate	no-Patv (0.881), no-Pab_mean (0.878), lagged-Patv (0.763), lagged pitch/wind-speed (0.684); all cross 0.65	Partial anchor robustness supported across all variants
Claim boundary	All 5-seed means cross 0.65; gate holds under single-anchor removal and dual-channel lagging	Use “partial anchor robustness” language, not anchor-free discovery

5.2 Boundary-Risk Vignette: Does the Gate Change Reserve Tradeoffs?

The reserve audit asks whether a recovered physical gate changes the reserve tradeoff in the operating window where over-forecasts become unmet reserve requirements. Four policies separate the effects: Graph WaveNet/global (low-RMSE reference), Graph WaveNet/physical-bin (physical stratification without learned gate), Boundary router/global (routed model with single reserve rule), and Boundary router/gate-bin (gate-conditioned allocation on the same routed predictor).

At shortage-to-reserve cost ratio 10, gate-conditioned binning exchanges about 3.6% additional reserve energy for a 14.5% reduction in shortage energy and a 1.38 percentage-point reduction in boundary-window violation relative to the same routed model with a global rule. The comparison is deliberately same-model: the reserve change is attributed to gate-conditioned allocation, not to a different forecasting backbone.

The validation-frozen quantile baseline changes how the result should be read. Boundary router/gate-bin remains better than the same routed model with a global quantile rule, but physical-bin quantile baselines are competitive and can be lower-cost in the boundary window. The strongest defensible statement is therefore attributional: the gate exposes a transition-window reserve-risk mechanism and improves a same-model global reserve rule, but it is not a universal reserve allocation policy.

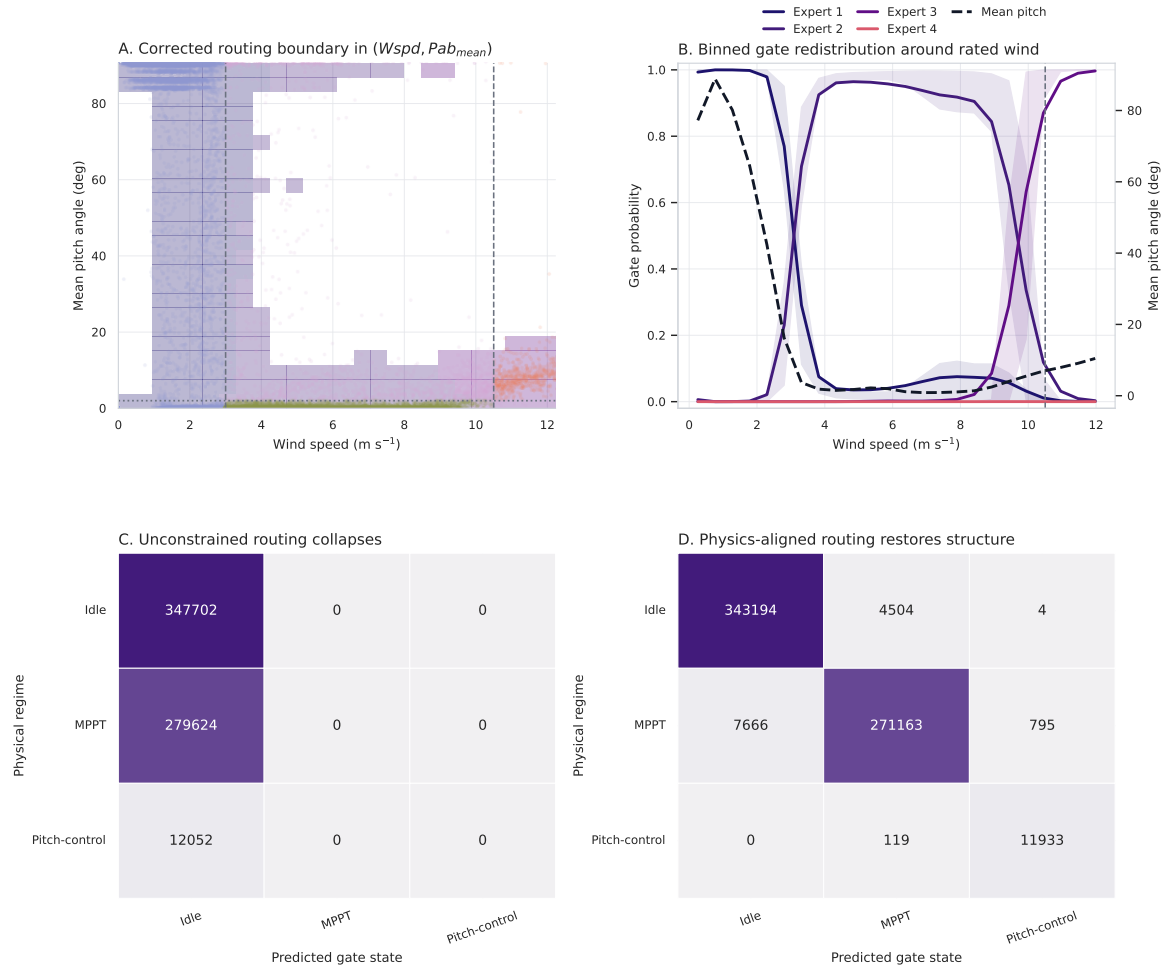


Figure 3: The WTB operating plane shows the main mechanism: after correction, the dominant routed responsibility changes around the rated-wind and pitch-control boundary instead of forming an arbitrary expert partition. The confusion matrices summarize the same recovery numerically.

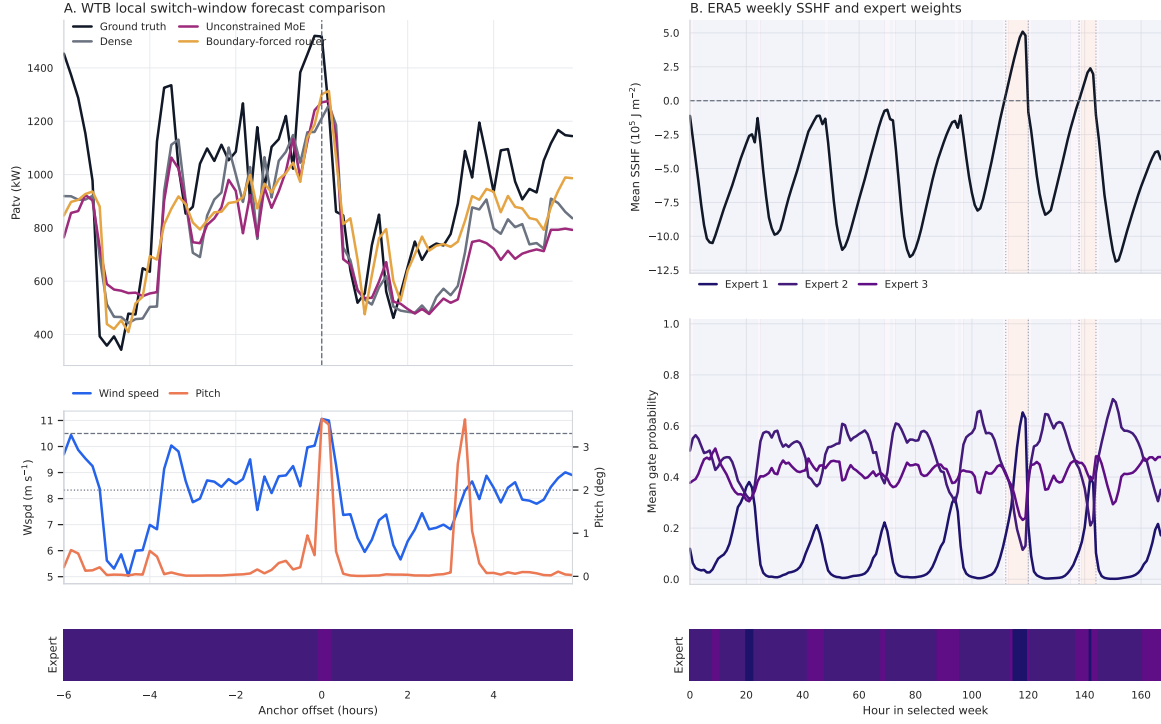


Figure 4: The time-series case studies show why observability matters. The WTB gate tracks a turbine-control switch that is partly hidden inside SCADA control action, whereas the ERA5 gate follows a more directly observed thermodynamic marker.

Table 5. Boundary-window reserve outcomes at shortage-to-reserve cost ratio 10.

Policy	Cost	Violation	Reserve	Shortage
Graph WaveNet/physical-bin	84.31M	0.0931	50.35M	3.40M
Boundary router/gate-bin	84.58M	0.0900	52.67M	3.19M
Boundary router/global	88.13M	0.1038	50.82M	3.73M
Graph WaveNet/global	88.80M	0.1022	50.32M	3.85M

Table 6. Validation-frozen boundary-window quantile reserve baselines at cost ratio 10.

Policy	Cost	Viol.	Reserve	Shortage	Reading
Boundary/physical-bin	83.78M	0.0880	53.26M	3.05M	Best same-router physical-bin reference
GWN/physical-bin	84.31M	0.0931	50.35M	3.40M	Low-RMSE backbone reference
Boundary/gate-bin	84.58M	0.0900	52.67M	3.19M	Gate diagnostic; same-model better than global
Boundary/global	88.13M	0.1038	50.82M	3.73M	Same-model global rule
GWN/global	88.80M	0.1022	50.32M	3.85M	Global low-RMSE reference

The useful reserve window is moderate rather than universal. At low shortage penalty, the validation-selected quantile tolerates too much residual shortage for gate-bin allocation to matter. At ratios 5–10, the gate lowers boundary shortage and violation while carrying more reserve. At ratio 20 the gain narrows, and at ratio 50 the same-model global rule is safer and cheaper. This pattern is exactly why the contribution is a reserve diagnostic: the gate identifies when a transition-window reserve rule has value and when the operator should use another rule.

Table 7. Boundary-window gate-bin value envelope relative to the same routed model with a global rule.

Ratio	Window	Cost	Viol.	Reserve	Shortage
2	inactive	+0.00M	+0.0000	+0.00M	+0.00M
5	moderate-cost window	-1.95M	-0.0170	+2.17M	-0.82M
10	moderate-cost window	-3.55M	-0.0138	+1.85M	-0.54M
20	narrow cost window	-1.39M	+0.0001	-0.70M	-0.03M
50	global rule favored	+5.93M	+0.0035	-1.04M	+0.14M

The toy operational-cost proxy is intentionally small: reserve procurement cost plus ρ times residual shortage energy. On the boundary slice the gate-bin rule reduces the same-model global proxy from 88.13M to 84.58M, while Graph WaveNet/physical-bin remains close at 84.31M. On the full sample Graph WaveNet/global is lower-cost, so the paper does not claim system-wide dispatch value. The operational consequence is bounded to the transition-window slice.

Table 8. Toy normalized operational-cost proxy at cost ratio 10.

Slice	Policy	Total	Reserve	Shortage
Boundary	Boundary/gate-bin	84.58M	52.67M	31.91M
Boundary	Boundary/global	88.13M	50.82M	37.31M
Boundary	GWN/physical-bin	84.31M	50.35M	33.96M
Full sample	GWN/global	464.07M	288.01M	176.06M
Full sample	Boundary/gate-bin	481.36M	291.16M	190.19M
Full sample	Boundary/global	501.32M	324.97M	176.35M

Horizon and time-of-day sensitivity checks, paired full-sample uncertainty, and operational-slice tables are kept as supplementary evidence.

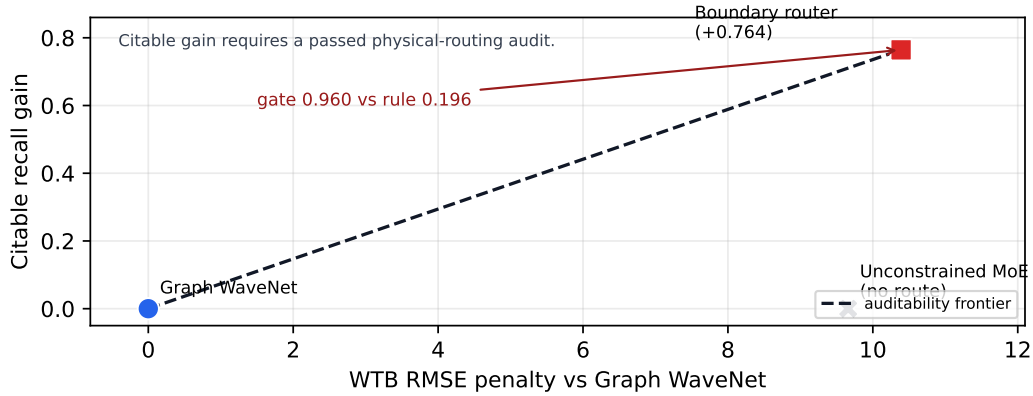


Figure 5: Accountability value versus forecasting RMSE price. Citable degraded-label gain is assigned only after the physical-routing audit passes.

5.3 What Are the Costs and Deployment Gates?

The accuracy cost is real and should be read as part of the design. Graph WaveNet and lag-feature baselines remain better whole-sample forecasters on WTB. The routed model is used when an operator or analyst needs an accountable operating-state assignment that can feed a boundary-specific decision, not when the sole target is minimum average error. Time-forward testing gives the same message in another form: late-period routing agreement remains high, but late-test RMSE rises sharply, showing that stable semantics do not guarantee stable value prediction under distribution shift.

The external wind-farm tests define the deployment gate. Kelmarsh/Penmanshiel checks do not pass the held-out routing criterion, identifying the conditions that must be verified before cross-farm use: pitch

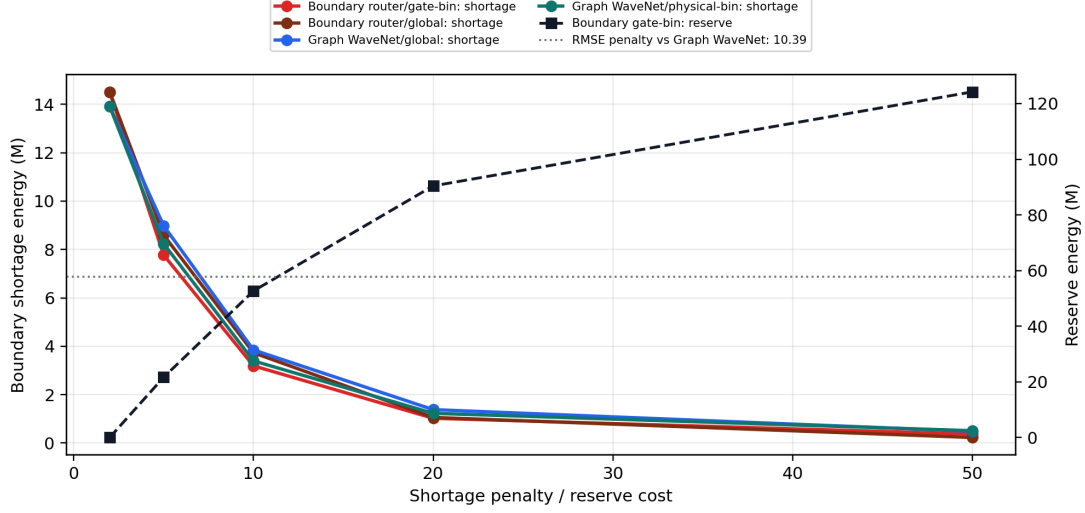


Figure 6: The decision curve places the reserve benefit beside the measured RMSE price. Gate-conditioned allocation is attractive only where the transition-window shortage reduction is worth the added reserve energy.

or proxy observability, boundary-cell support, compatible turbine geometry, local threshold estimation, and a held-out routing pass. The method exports an evidence protocol, not a promise of automatic transfer.

Table 9. External-site deployment gates identified by Kelmarsh/Penmanshiel testing.

Gate	Observed evidence	Required decision before reserve use
Held-out routing criterion	Default WTB boundary gives mean external NMI 0.4877, below the 0.50 criterion	Re-establish routing agreement locally
Local boundary recalibration	Average test NMI changes from 0.1324 to 0.1491 after validation-only recalibration	Treat threshold transfer as insufficient
Pitch/proxy observability	One transfer direction has no effective pitch-feature coverage or boundary cells	Require pitch channels or a validated proxy
Calibration-window adaptation	Small-window adaptation remains below the held-out balanced-accuracy threshold	Use calibration only after a held-out pass
Geometry and regime support	Farm scale, sensor fields, power curves, and regime shares differ across sites	Run a local evidence protocol before gate-bin reserve allocation

The WTB internal proxy drill shows the minimum evidence sequence: verify sensor coverage, re-estimate local boundary and gate-to-regime map, pass held-out routing, then run the reserve diagnostic. Stronger gate repair is justified only when the reserve-window benefit exceeds the measured RMSE and reserve-energy cost.

6 Limitations

The WTB correction depends on threshold-based pseudo-labels derived from wind speed and mean pitch angle. Those same channels also appear in the gate anchor, so the method deliberately constrains the router to a declared operating boundary. The label-degradation audit shows value under delayed, missing, or noisy labels, but it should not be read as anchor-free physical discovery or as a general future-label

New wind-farm local evidence protocol: physical interpretation requires every check to pass

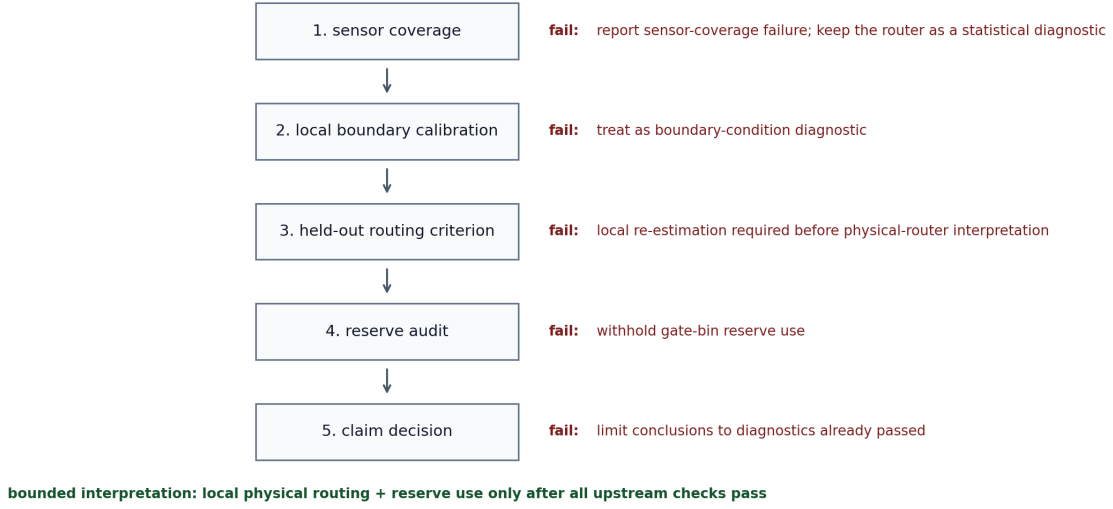


Figure 7: The new-site protocol turns external failure into a practical safety check: sensor coverage, local boundary calibration, held-out routing, and reserve audit must all pass before gate-conditioned reserve allocation is used.

switch predictor. The claim is boundary-aware regularized routing that remains useful when the operational label stream degrades.

The active-power anchor has a strict timing requirement. Pat_v is allowed only as the historical/anchor-time active-power measurement available when the forecast is issued; future active power remains the supervised target. This is a defensible SCADA forecasting convention, but it is also a portability constraint. The five-seed anchor-stress evidence (Table 7) shows that removing Pat_v or Pab_mean individually leaves the gate essentially intact, while lagging both channels lowers the mean NMI but still crosses the 0.65 threshold. If a deployment environment delays active-power telemetry or changes channel definitions, the Pat_v anchor must be removed or lagged and the leakage, intervention, and reserve diagnostics must be rerun.

The expert-regime language is tied to the implemented logit convention. The first primary-regime logits are fixed to the operating labels during supervised alignment, which prevents seed-wise permutation for those anchored meanings. That makes cross-seed MPPT/pitch statements interpretable, but only for the anchored logits and only under the declared mapping. Unassigned experts and wake auxiliary logits should not be overread as universal turbine states.

The comparison is bounded. WTB includes graph, transformer, lag-feature, persistence, power-curve, DLinear-style references, and validation-frozen empirical quantile reserve baselines, while ERA5 includes persistence and several learned baselines. Larger time-series backbones, broader graph-transformer variants, trained quantile-regression forecasters, scenario reserve, distributional reserve, and security-constrained reserve methods are outside the present benchmark.

The reserve audit is a normalized proxy: fixed empirical quantile bins calibrated on validation shortfall and tested once on held-out predictions. It ranks boundary-window tradeoffs under declared cost ratios, but it is not a market-price or security-constrained dispatch study. External diagnostics fail the held-out routing criterion; Kelmarsh/Penmanshiel testing identifies the sensor coverage, pitch observability, local boundary re-estimation, and held-out routing checks that must pass before cross-farm use. The time-forward audit shows that late-period distribution shift preserves routing agreement while degrading

value prediction: the method supports routing semantics, not time-stable forecasting accuracy.

Statistical reliability is uneven (Supplementary Table A6). The strongest claims are the five-seed WTB routing-recovery, mechanism-intervention, and within-WTB spatial/temporal stress results; gate-alignment deltas versus the full MoE and boundary-window reserve-cost comparisons remain bounded diagnostics, not superiority claims.

7 Conclusion

This paper introduces SCADA-anchored regime-aware routing as an auditable control-boundary framework for wind-turbine forecasting. The boundary-forced router recovers the declared WTB MPPT-to-pitch partition (NMI about 0.87, ARI about 0.92), keeps 0.960 early pitch-window recall under a six-step label delay while the delayed threshold rule falls to 0.196, and exposes transition-window reserve risk. This accountability has a measured accuracy price: overall RMSE 236.13 versus 225.74 for Graph WaveNet. Validation-frozen quantile baselines, Supplementary Table A6, and Kelmarsh/Penmanshiel deployment gates keep the claim bounded. The contribution is not forecasting SOTA or universal reserve-policy optimality, but an auditable MPPT-to-pitch routing tool with degraded-label value, RMSE price, and explicit transfer conditions.

8 Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI ChatGPT/Codex to support language editing, consistency checking, and submission-material drafting. The tools were not used to generate data, run analyses, create references, or determine scientific conclusions. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

9 Code and data availability

The raw datasets are publicly available: KDD Cup 2022, Kelmarsh, and Penmanshiel SCADA records. Raw third-party data are not redistributed. Code, configuration files, releasable derived tables, figure data, and model checkpoints will be made available with the article. The reproduction package includes end-to-end scripts for the WTB routing analysis, reserve audit, anchor-stress cache/train/guard protocol, anchor-stress early-warning label-degradation audit, and external boundary diagnostics. The provenance package lists the anchor-stress cache/train/guard and early-warning protocols as mandatory. The analysis involves no human subjects.

References

- [1] Wu, Z., Pan, S., Long, G., Jiang, J., Zhang, C., *Graph WaveNet for deep spatial-temporal graph modeling*, *Proceedings of the 28th international joint conference on artificial intelligence* 1907–1913;2019, doi:10.24963/ijcai.2019/264.
- [2] Park, J., Park, J., Physics-induced graph neural network: An application to wind-farm power estimation, *Energy* 187, 115883;2019, doi:10.1016/j.energy.2019.115883.
- [3] Yu, M., Zhang, Z., Li, X., Yu, J., Gao, J., Liu, Z., You, B., Zheng, X., Yu, R., Superposition graph neural network for offshore wind power prediction, *Future Generation Computer Systems* 113, 145–157;2020, doi:10.1016/j.future.2020.06.024.
- [4] Kim, D., Ryu, G., Moon, C., Accuracy of a short-term wind power forecasting model based on deep learning using LiDAR-SCADA integration: A case study of the 400-MW Anholt offshore wind farm, *Applied Energy* 373, 123882;2024, doi:10.1016/j.apenergy.2024.123882.
- [5] Daenens, S., Verstraeten, T., Daems, P.-J., Nowé, A., Helsen, J., Spatio-temporal graph neural networks for power prediction in offshore wind farms using SCADA data, *Wind Energy Science* 10, 1137–1152;2025, doi:10.5194/wes-10-1137-2025.
- [6] Jacobs, R.A., Jordan, M.I., Nowlan, S.J., Hinton, G.E., Adaptive mixtures of local experts, *Neural Computation* 3, 79–87;1991, doi:10.1162/neco.1991.3.1.79.
- [7] Jordan, M.I., Jacobs, R.A., Hierarchical mixtures of experts and the EM algorithm, *Neural Computation* 6, 181–214;1994, doi:10.1162/neco.1994.6.2.181.
- [8] Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., Dean, J., *Outrageously large neural networks: The sparsely-gated mixture-of-experts layer*, *International conference on learning representations* 2017, <https://openreview.net/forum?id=BlckMDqlg>.
- [9] Fedus, W., Zoph, B., Shazeer, N., Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity, *Journal of Machine Learning Research* 23, 1–39;2022, <https://www.jmlr.org/papers/v23/21-0998.html>.
- [10] Shi, S., Han, L., Liu, Y., others, *Time-MoE: Billion-scale time series foundation models with mixture of experts*, *International conference on learning representations* 2025, <https://arxiv.org/abs/2409.16040>.
- [11] Pinson, P., Wind energy: Forecasting challenges for its operational management, *Statistical Science* 28, 564–585;2013, doi:10.1214/13-STS445.
- [12] Wang, Y., Zhang, F., Kou, H., Zou, R., Hu, Q., Wang, J., A review of predictive uncertainty modeling techniques and evaluation metrics in probabilistic wind speed and wind power forecasting, *Applied Energy* 396, 126234;2025, doi:10.1016/j.apenergy.2025.126234.
- [13] Guo, S., Lin, Y., Feng, N., Song, C., Wan, H., *Attention based spatial-temporal graph convolutional networks for traffic flow forecasting*, *Proceedings of the AAAI conference on artificial intelligence* 33, 922–929;2019, doi:10.1609/aaai.v33i01.3301922.
- [14] Bai, L., Yao, L., Li, C., Wang, X., Wang, C., *Adaptive graph convolutional recurrent network for traffic forecasting*, *Advances in neural information processing systems* 33, 17804–17815;2020.
- [15] Karpatne, A., Atluri, G., Faghmous, J.H., Steinbach, M., Banerjee, A., Ganguly, A., Shekhar, S., Samatova, N.F., Kumar, V., Theory-guided data science: A new paradigm for scientific discovery from data, *IEEE Transactions on Knowledge and Data Engineering* 29, 2318–2331;2017, doi:10.1109/TKDE.2017.2720168.
- [16] Karniadakis, G.E., Kevrekidis, I.G., Lu, L., Perdikaris, P., Wang, S., Yang, L., Physics-informed machine learning, *Nature Reviews Physics* 3, 422–440;2021, doi:10.1038/s42254-021-00314-5.
- [17] Parsa, S.M., Physics-informed machine learning meets renewable energy systems: A review of advances, challenges, guidelines, and future outlooks, *Applied Energy* 402, 126925;2025, doi:10.1016/j.apenergy.2025.126925.
- [18] Tautz-Weinert, J., Watson, S.J., Using SCADA data for wind turbine condition monitoring: A review, *IET Renewable Power Generation* 11, 382–394;2017, doi:10.1049/iet-rpg.2016.0248.

- [19] Zhou, J., Lu, X., Xiao, Y., Tang, J., Su, J., Li, Y., Liu, J., Lyu, J., Ma, Y., Dou, D., SDWPF: A dataset for spatial dynamic wind power forecasting over a large turbine array, *Scientific Data* 11, 649;2024, doi:10.1038/s41597-024-03427-5.
- [20] Doherty, R., O'Malley, M., A new approach to quantify reserve demand in systems with significant installed wind capacity, *IEEE Transactions on Power Systems* 20, 587–595;2005, doi:10.1109/TPWRS.2005.846206.
- [21] Ela, E., Milligan, M., Kirby, B., *Operating reserves and variable generation*, 2011, doi:10.2172/1023095, <https://www.osti.gov/biblio/1023095>.
- [22] Bremnes, J.B., Probabilistic wind power forecasts using local quantile regression, *Wind Energy* 7, 47–54;2004, doi:10.1002/we.107.
- [23] Zhang, Y., Wang, J., Wang, X., Review on probabilistic forecasting of wind power generation, *Renewable and Sustainable Energy Reviews* 32, 255–270;2014, doi:10.1016/j.rser.2014.01.033.
- [24] Hersbach, H., Bell, B., Berrisford, P., Hirahara, S., Horányi, A., Muñoz-Sabater, J., Nicolas, J., Peubey, C., Radu, R., Schepers, D., Simmons, A., Soci, C., Abdalla, S., Abellan, X., Balsamo, G., Bechtold, P., Biavati, G., Bidlot, J.-R., Bonavita, M., De Chiara, G., Dahlgren, P., Dee, D., Diamantakis, M., Dragani, R., Flemming, J., Forbes, R., Fuentes, M., Geer, A., Haimberger, L., Healy, S., Hogan, R., Hólm, E., Janisková, M., Keeley, S., Laloyaux, P., Lopez, P., Lupu, C., Radnoti, G., Rosnay, P. de, Rozum, I., Vamborg, F., Villaume, S., Thépaut, J.-N., The ERA5 global reanalysis, *Quarterly Journal of the Royal Meteorological Society* 146, 1999–2049;2020, doi:10.1002/qj.3803.

Appendix A. Training and implementation details

Training loop summary

Algorithm A1 Physics-aligned MoE training

Input: history windows X , graph sequence A , targets Y , primary regime anchors R , auxiliary wake labels W (WTB only), model parameters θ

for each minibatch do

 Build node-level graph context from the precomputed graph tensors

 Encode X with the directed-diffusion GRU to obtain node context H

 Compute gate logits Z and soft routing probabilities G

 Compute expert forecasts and aggregate them with G

 Evaluate L_{pred}

 If MoE mode is active:

 add L_{bal} on top- K routing statistics

 add L_{align} on valid primary regime anchors

 if WTB: add L_{force} on MPPT/pitch samples

 if WTB: add L_{aux} on wake labels

 add L_{smooth} on the final graph snapshot

 Update θ with AdamW and mixed precision

end for

Select the checkpoint with the best validation RMSE

Model-family and supplementary evidence index

Table A1. In-family model comparison used for mechanism validation.

Model	Routing	Extra terms	Params	Role
Dense (matched)	Single dense head	None	WTB 110,012 / ERA5 103,272	Shared-mapping reference
Unconstrained MoE	Node-level soft gate	Prediction loss only	WTB 110,012 / ERA5 103,843	Routed-capacity control
Corrected routing comparator	Node-level soft gate	WTB boundary-forced: $L_{\text{bal}} + L_{\text{align}} +$ L_{force} ; ERA5: full corrected stack	WTB 110,012 / ERA5 103,843	Accountability comparator

Auxiliary losses

Let $\mathcal{B}$ denote the routed samples after masks, with $B = |\mathcal{B}|$. The soft importance and normalized share of expert e are $I_e = \frac{1}{B} \sum_{n=1}^B g_n^{(e)}$ and $P_e = I_e / \sum_r I_r$. The top- K load is $f_e = \frac{1}{B} \sum_{n=1}^B \mathbf{1}[e \in \text{TopK}(\mathbf{g}_n)]$, giving

$$\mathcal{L}_{\text{bal}} = E \sum_{e=1}^E f_e P_e - 1.$$

For WTB wake supervision,

$$\mathcal{L}_{\text{aux}} = \frac{1}{|\Omega_{\text{wake}}|} \sum_{(i,t) \in \Omega_{\text{wake}}} \text{BCE}(z_{i,t}^{(\text{wake})}, W_{i,t}),$$

where Ω_{wake} contains only MPPT and pitch-control samples with defined wake flags. The graph-smoothness penalty is

$$\mathcal{L}_{\text{smooth}} = \frac{\sum_{t \in \mathcal{T}} \sum_{i,j \in \mathcal{B}} \mathcal{A}_t(i,j) \|\mathbf{g}_{i,t} - \mathbf{g}_{j,t}\|_2^2}{\sum_{t \in \mathcal{T}} \sum_{i,j \in \mathcal{B}} \mathcal{A}_t(i,j) + \epsilon}.$$

Graph construction details

For WTB, the downstream unit vector is $\mathbf{u}_{i,t} = [\sin(\theta_{i,t} + \pi), \cos(\theta_{i,t} + \pi)]^\top$. With $\Delta \mathbf{p}_{ij} = \mathbf{p}_j - \mathbf{p}_i$, the streamwise and cross-stream distances are $d_{ij,t}^\parallel = \Delta \mathbf{p}_{ij}^\top \mathbf{u}_{i,t}$ and $d_{ij,t}^\perp = |\Delta p_{ij}^x u_{i,t}^y - \Delta p_{ij}^y u_{i,t}^x|$. A candidate edge activates when

$$\mathbb{I}_{ij,t}^{\text{cone}} = \mathbf{1} \left[d_{ij,t}^\parallel > 0 \wedge \arctan \left(\frac{d_{ij,t}^\perp}{\max(d_{ij,t}^\parallel, 10^{-6})} \right) \leq \phi \right].$$

The wake weight is $\tilde{\mathcal{A}}_t(i,j) = \exp(-d_{ij,t}^\parallel / \alpha) \exp(-|d_{ij,t}^\perp| / \beta) \mathbb{I}_{ij,t}^{\text{cone}}$. If wind direction is missing, fall back to $\mathcal{A}^{\text{static}}(i,j) = \exp(-\|\Delta \mathbf{p}_{ij}\|_2 / d_{\text{max}})$. Only the strongest M inbound weights are retained. The wake score and flag are $s_{i,t}^{\text{wake}} = \sum_j \mathcal{A}_t(i,j)$ and

$$W_{i,t} = \begin{cases} 1, & s_{i,t}^{\text{wake}} \geq q_{0.75}^{\text{wake}} \wedge R_{i,t}^{\text{wtb}} \in \{1, 2\}, \\ 0, & s_{i,t}^{\text{wake}} < q_{0.75}^{\text{wake}} \wedge R_{i,t}^{\text{wtb}} \in \{1, 2\}, \\ \emptyset, & \text{otherwise.} \end{cases}$$

For ERA5, the retained symmetric graph uses a Gaussian kernel on great-circle distances:

$$\mathcal{A}(i,j) = \exp \left(-\frac{d_{ij}^2}{2\sigma^2} \right) \mathbf{1}[j \in \mathcal{N}_{k_{\text{nn}}}(i) \text{ or } i \in \mathcal{N}_{k_{\text{nn}}}(j)],$$

where σ is the median retained neighbor distance on the training graph.

Shared constants

Table A3. Shared architecture, graph, and training constants used in the reported experiments.

Item	WTB	ERA5	Role
History / horizon (H, P)	(36, 24)	(36, 24)	Input and forecast lengths
Gate softmax temperature τ	0.7	0.7	Soft routing sharpness
Top- K in L_{bal}	1	1	Balancing statistic only
Hidden size d	64	64	Shared encoder width
k_{nn} candidates	8	8	Initial neighborhood size
Maximum candidate radius d_{max}	1500 m	not used	WTB proximity fallback / candidate filter
Wake-cone half-angle ϕ	25°	not used	WTB downstream activation
Streamwise decay α	1200 m	not used	WTB wake attenuation
Cross-stream decay β	400 m	not used	WTB wake attenuation
Retained inbound edges M	5	symmetric k_{nn} graph	Final graph sparsity
Optimizer	AdamW	AdamW	Shared optimizer
Learning rate	2×10^{-3}	2×10^{-3}	Shared schedule
Weight decay	10^{-4}	10^{-4}	Shared regularization
Batch size	16	16	Shared mini-batch size
Gradient clipping	1.0	1.0	Shared training stabilization

Dataset-specific thresholds and routing weights

Table A4. Dataset-specific regime thresholds used to construct routing anchors.

Dataset	Threshold	Value and meaning
WTB	u_{idle}	3.0 m s ⁻¹ , idle / active split
WTB	u_{rated}	10.5 m s ⁻¹ , MPPT / pitch boundary
WTB	p_{th}	2.0 deg, pitch-angle threshold
WTB	$q_{0.75}^{\text{wake}}$	1.0690, upper-quartile wake-score threshold
ERA5	$\varepsilon_{\text{sshf}}$	169,116, 10th percentile of $ \text{sshf} $ on the training split
ERA5	$q_{0.95}^{\Delta}$	458,287.125, 95th percentile of $ \Delta \text{sshf} $ on the training split

Table A5. Active routing-loss weights in the reported corrected models.

Weight	WTB	ERA5	Role
λ_{bal}	1000	0.01	Expert-usage stabilization
λ_{align}	5000	0.2	Coarse regime alignment
λ_{force}	10000	0	Boundary-focused forcing
λ_{aux}	250	0	Wake-sensitive auxiliary routing
λ_{smooth}	0.05	0.05	Local routing coherence

Statistical claim boundaries

Table A6 records the reviewer-facing statistical boundary used for wording. The RMSE price versus Graph WaveNet is FDR-significant, gate-alignment deltas versus the full physics-aligned MoE are positive but not FDR-significant, and boundary-window quantile reserve comparisons have bootstrap intervals crossing zero. These tests are why the main text uses price, diagnostic, and bounded-claim language rather than forecast- or reserve-superiority wording.

Table A6. Statistical claim boundaries used for reviewer-facing wording.

Claim	Estimate	95% CI	p_{BH}	Wording consequence
Accuracy price vs Graph WaveNet	9.029	–	<0.001	Boundary router is significantly worse on headline RMSE
Boundary-band accuracy price	17.758	–	0.001	The price also appears in the boundary window
Gate NMI vs full physics-aligned MoE	0.040	[0.008, 0.071]	0.066	Positive but not FDR-significant; cite as bounded mechanism contrast
Gate ARI vs full physics-aligned MoE	0.035	[-0.000, 0.078]	0.126	Positive but not FDR-significant; avoid superiority wording
Boundary quantile cost vs GWN physical bin	-0.263M	[-10.704M, 9.782M]	–	CI crosses zero; reserve cost should remain a diagnostic claim
Boundary quantile violation vs GWN physical bin	0.003	[-0.011, 0.020]	–	CI crosses zero; no universal reserve-policy optimality claim